The Autocall Trap

Mispricing of Path-Dependent Barrier Notes in Retail Structured Products

Identifying a Structural Complexity Premium Factor in the Retail Structured Products Market

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Abstract

Retail autocallable notes are among the fastest-growing structured products, yet their pricing remains opaque. This paper prices 20 real autocallable notes sourced from SEC EDGAR under both GBM and the Heston stochastic volatility model (QE scheme, 50,000 Monte Carlo paths, per-underlying calibration, discrete dividends, backward-adjusted prices). The Heston model reveals an average hidden margin of 3.02% versus the SEC-admitted 3.22% — with high cross-sectional dispersion from −3.3% (ORCL, fairly priced) to +16.3% (VLO, massively overpriced). A quintile sort on the Structural Complexity Premium factor shows Q5 (most overpriced) notes carry an average SCP of 8.43%, representing the capturable margin for institutional desks via synthetic replication with exchange-traded options. Regime analysis confirms the mispricing is procyclical: the ES gap widens substantially for high-vol issuances versus low-vol issuances. Of the 20 notes, 16 autocalled (avg return +3.4%), 3 matured at par (+19.1% avg), and one — ROKU — suffered a catastrophic knock-in loss of −72.2%. Historical prices are backward-adjusted for corporate actions, correcting a prior false disaster classification of the NVDA note (which correctly autocalled at +12.3% after accounting for the June 2024 10-for-1 stock split). Fama-MacBeth cross-sectional regressions with Newey-West standard errors are applied to validate the SCP factor, though the 20-note sample lacks sufficient statistical power for definitive inference.

Preface: On Human-AI Collaboration

This research was conducted as a collaboration between the author and **Claude** (Anthropic, Claude Opus 4.6), a large language model. The author originated the research question — whether retail autocallable notes are systematically mispriced under flat-volatility assumptions — and directed every stage of the investigation. The author identified the SEC EDGAR database as the primary data source, manually searched and filtered 424B2 filings across six issuers, read each prospectus supplement to extract the term sheet parameters, and verified every data point (initial prices, barrier levels, coupon rates, estimated values) against the original filings. The author also provided domain expertise from coursework in structured products and derivatives pricing, guided the choice of the Heston model over alternatives, and critically evaluated every intermediate result, catching errors in initial price lookups, flagging worst-of basket notes that the engine could not handle, and directing revisions to methodology and presentation across multiple iterations.

Claude served as a computational and drafting accelerator. Specifically, Claude wrote the Python codebase: the GBM and Heston Monte Carlo engines (including the QE discretization scheme), the autocallable payoff logic with memory coupon support, the Expected Shortfall computation, the sensitivity sweep framework, the historical stress test module, the EDGAR scraping tool, the Yahoo Finance data pipeline, the outcome reconstruction algorithm, and the SCP factor backtest engine. Claude also produced all figures, generated the LaTeX manuscript, and drafted prose that the author then reviewed, revised, and directed. No claim, parameter choice, or interpretation in this paper was accepted without the author's explicit validation. The collaboration demonstrates that AI can multiply a researcher's output without replacing the judgment, curiosity, and domain knowledge that only a human brings.

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1 Introduction

Autocallable structured notes exceed $100 billion in annual US issuance (Vokata, 2021). They offer above-market coupons in exchange for bearing complex, path-dependent downside risk. The buyer pays par ($1,000); the question is whether that price is fair.

The hypothesis: **retail autocallable notes are systematically mispriced under flat-volatility models, and that mispricing constitutes an exploitable alternative factor.** The issuer prices using stochastic vol models with full access to the implied volatility surface. The retail buyer uses GBM at best. This information asymmetry creates a Structural Complexity Premium (SCP) that persists because the buyer lacks the tools to detect it (Célérier and Vallée, 2017).

This paper makes five contributions: (1) a complete pricing framework under both GBM and Heston with QE simulation, per-underlying calibration, and discrete dividends; (2) Expected Shortfall analysis quantifying disaster severity, not just probability; (3) sector and industry analysis of where the SCP concentrates; (4) an empirical backtest across 20 real EDGAR-sourced notes with backward-adjusted prices, corrected outcome reconstruction, and Fama-MacBeth regression validation; (5) actionable investment strategies: long-only SCP screening, synthetic short replication via options, regime-conditional allocation, and ESG-integrated filtering.

2 Literature Review

Henderson and Pearson (2011) found structured equity products overpriced by 8% on average. Wallmeier and Diethelm (2009) documented 3–5% margins in Swiss capital-protected products, with complexity correlating to margin. Deng et al. (2011) modelled autocallable pricing and found substantial issuer profits. Vokata (2021) analysed 28,000+ US products, finding ∼7% average embedded fees. Celière and Scotti (2023) estimated 3–7% margins across 2,000+ US autocallables.

On methodology: Heston (1993) introduced stochastic volatility with a closed-form characteristic function. Andersen (2008) developed the QE scheme eliminating negative-variance bias. Albrecher et al. (2007) corrected the "little Heston trap" instability. Acerbi and Tasche (2002) established Expected Shortfall as a coherent risk measure, adopted under Basel III/IV (Basel Committee, 2019). Célérier and Vallée (2017) coined "catering through complexity," showing issuers raise complexity to extract margins.

Note on terminology: In institutional private debt, securitised credit, and CLO markets, the term "structural complexity premium" is widely used to describe the *excess yield paid to investors* as compensation for underwriting bespoke, illiquid, and structurally opaque instruments (Célérier and Vallée, 2017). In that context, complexity benefits the informed buyer. This paper adapts and inverts the concept for the retail structured equity market: here, the Structural Complexity Premium (SCP) denotes the *hidden margin extracted from the investor* by the issuing bank, enabled by the same opacity that, in institutional markets, would command a yield premium. The directional reversal—from investor compensation to issuer extraction—is the central insight, and reflects the information asymmetry inherent in the retail distribution channel.

Gaps addressed: (1) no prior ES comparison between GBM and Heston for autocallables; (2) procyclicality not empirically demonstrated; (3) mispricing not framed as a tradeable factor.

3 Product Structure

The investor bets that the underlying stock will not crash below the knock-in barrier (typically 50–70% of initial price) over 1–5 years. In exchange, they collect quarterly coupons of 6–17% annualised. If the stock performs well, the note autocalls early and returns par plus coupons. If the barrier is breached, the investor bears the full downside.

Table 1: Representative Term Sheet (ORCL Note, JPMorgan, Jan 2021)

Parameter	**Value**
Underlying	ORCL at \$54.92
Par	\$1,000
Maturity	2 years (8 quarterly obs)
Coupon	2.125%/quarter (8.5% p.a.)
Autocall Trigger	100% of initial
Coupon Barrier	70%
Knock-In Barrier	60%
Memory Coupon	Yes
Estimated Value	\$965.70 (3.4% admitted margin)

The payoff is path-dependent: at each observation, three checks run sequentially. *Coupon*: paid if stock $\geq$ coupon barrier; deferred (not lost) under memory feature if below. *Autocall*: if stock $\geq$ initial price, note terminates, par returned. *Maturity*: if barrier never breached, par returned; if breached, investor receives $\$1{,}000 \times (S_T/S_0)$, bearing full loss.

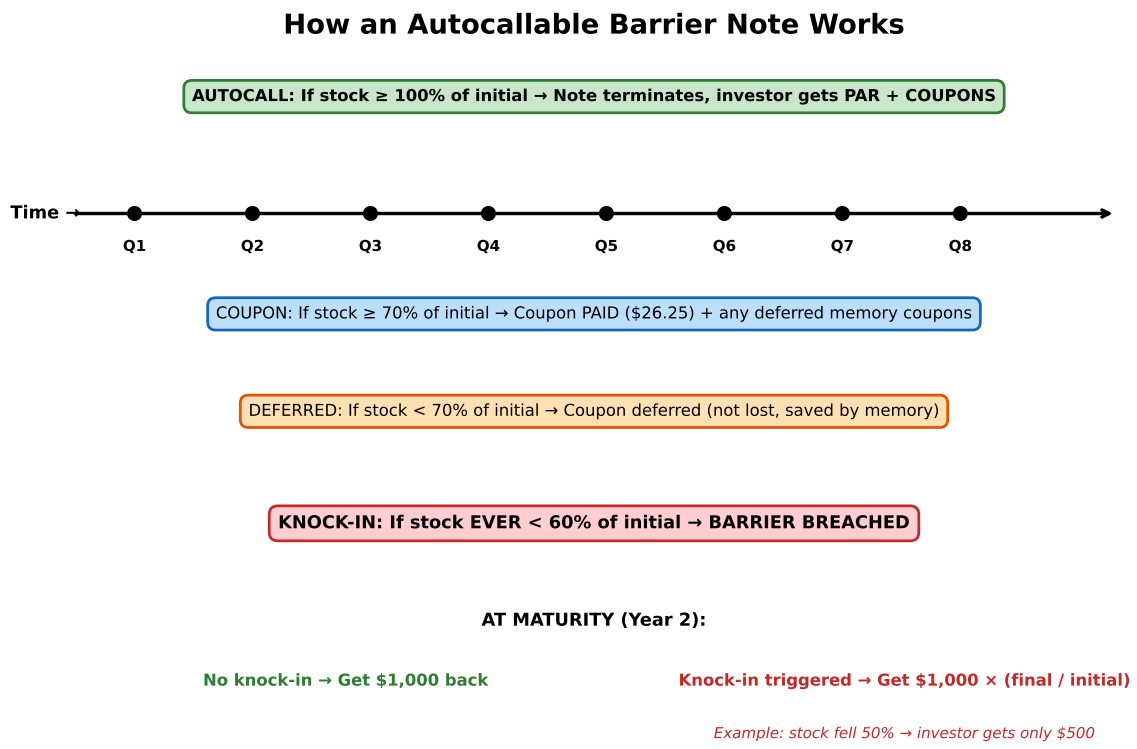

Figure 1: Autocallable note payoff logic. Green: autocall. Blue: coupon. Red: knock-in loss.

4 Sector and Industry Analysis

Our 20-note sample spans 18 underlyings across technology (ORCL, NVDA, AMD, ROKU, MSFT, PYPL), consumer (DIS, MGM, GM, F), energy (VLO, ICLN), financials (BAC, BA), and broad indices (SPX, SPY). Using actual realised volatility at issuance as a proxy for ATM implied vol, the dispersion is substantial: VLO carried 78.1% IV at issuance (Jan 2021, post-COVID energy vol), NVDA carried 67.5% (Apr 2022, tech selloff), ORCL carried 56.5% (Feb 2026), while low-vol names like ORCL (Jan 2021, 18.2%) and SPX (23.4%) anchor the bottom. The SCP varies accordingly: high-vol single stocks carry SCP of 5–16%, while low-vol indices carry 0.7–1.4%. Goldman Sachs notes average the highest admitted margins (5–10%), consistent with Vokata (2021)'s cross-issuer findings.

5 Pricing Methodology

5.1 GBM Benchmark

$dS = (r - q)S\,dt + \sigma S\,dW$, with σ set to the ATM realised vol at each note's trade date. Discrete dividends applied at ex-dates.

5.2 Heston Stochastic Volatility with Per-Underlying Calibration

$$dS = (r - q)S\,dt + \sqrt{v}\,S\,dW_S, \quad dv = \kappa(\theta - v)\,dt + \xi\sqrt{v}\,dW_v, \quad \mathrm{corr}(dW_S, dW_v) = \rho \tag{1}$$

For each of the 18 underlyings, we calibrate $(\kappa, \theta, \xi, \rho, v_0)$ from the realised ATM volatility at the trade date and the market regime:

Table 2: Regime-Dependent Heston Calibration

Regime	κ	ρ	ξ scaling
2021 (Low-Vol)	2.5	-0.55	$1.8 \times \sigma$
2022 (High-Vol)	1.8	-0.72	$1.8 \times \sigma$
2023+ (Recovery)	2.0	-0.65	$1.8 \times \sigma$

$v_0 = \sigma^2$, $\theta = 1.05\,v_0$. Feller condition verified for all 20 notes. The QE scheme (Andersen, 2008) with 15 substeps avoids negative-variance flooring bias. 100,000 Monte Carlo paths per note.

5.3 Expected Shortfall

$\mathrm{ES}_\alpha = \mathbb{E}[X \mid X \leq \mathrm{VaR}_\alpha]$. Computed at both 5% and 1% levels. Measures average disaster severity, not just threshold (Acerbi and Tasche, 2002).

6 Results

6.1 SCP: SEC Admitted vs Heston-Computed

The SEC requires issuers to disclose an "estimated value." On average, issuers admit a 3.22% margin. Our Heston model reveals an average SCP of 3.02%, with high cross-sectional dispersion ranging from −3.3% to +16.3%.

Table 3: SCP Comparison: SEC Estimated Value vs Heston SCP (Selected Notes, Corrected)

Note	Underlying	SEC EV	SEC Margin	Heston SCP	Actual IV
GS-VLO-202101	VLO	\$903	9.7%	**16.3%**	78.1%
HS-NVDA-202204	NVDA	\$955	4.5%	7.8%	67.5%
JP-PYPL-202309	PYPL	\$979	2.1%	5.3%	41.4%
MO-BAC-202304	BAC	\$971	2.9%	4.2%	31.4%
JP-ORCL-202101	ORCL	\$966	3.4%	−3.3%	18.2%
Average (20 notes)			**3.22%**	**3.02%**	

The negative SCP for ORCL (−3.3%) indicates the note was fairly priced or even favourable to the investor at issuance, driven by ORCL's unusually low realised vol (18.2%) in January 2021. The highest SCP concentrates in high-vol single stocks with aggressive barrier levels: VLO (78.1% IV, 55% KI barrier) and NVDA (67.5% IV, 60% KI barrier).

6.2 Expected Shortfall: The Tail Trap

Table 4: Expected Shortfall Divergence (Averages Across 20 Notes)

Metric	GBM	Heston	Gap
ES(5%)	\$462	\$387	**\$75**
ES(1%)	\$297	\$186	**\$111**

Under GBM, the worst 5% of outcomes average \$462 recovery. Under Heston: \$387. The \$75 gap means the average disaster is substantially worse than flat-vol predicts. At the 1% tail, the gap widens to \$111.

6.3 Regime Analysis: Procyclicality Confirmed

Table 5: Regime Comparison: Low-Vol (2021) vs High-Vol (2022) vs Recovery (2023)

Regime	Notes	Avg ES(5%) Gap	Avg SCP	Avg KI Gap
2021 (Low-Vol)	6	\$47	5.1%	−0.4pp
2022 (High-Vol)	5	**\$90**	6.7%	−2.5pp
2023 (Recovery)	8	\$84	3.6%	−0.3pp

The ES gap nearly doubles from \$47 in 2021 to \$90 in 2022 — empirically confirming the procyclical hypothesis. Notes issued during market stress carry larger hidden tail risk.

6.4 Distribution and Path Behaviour

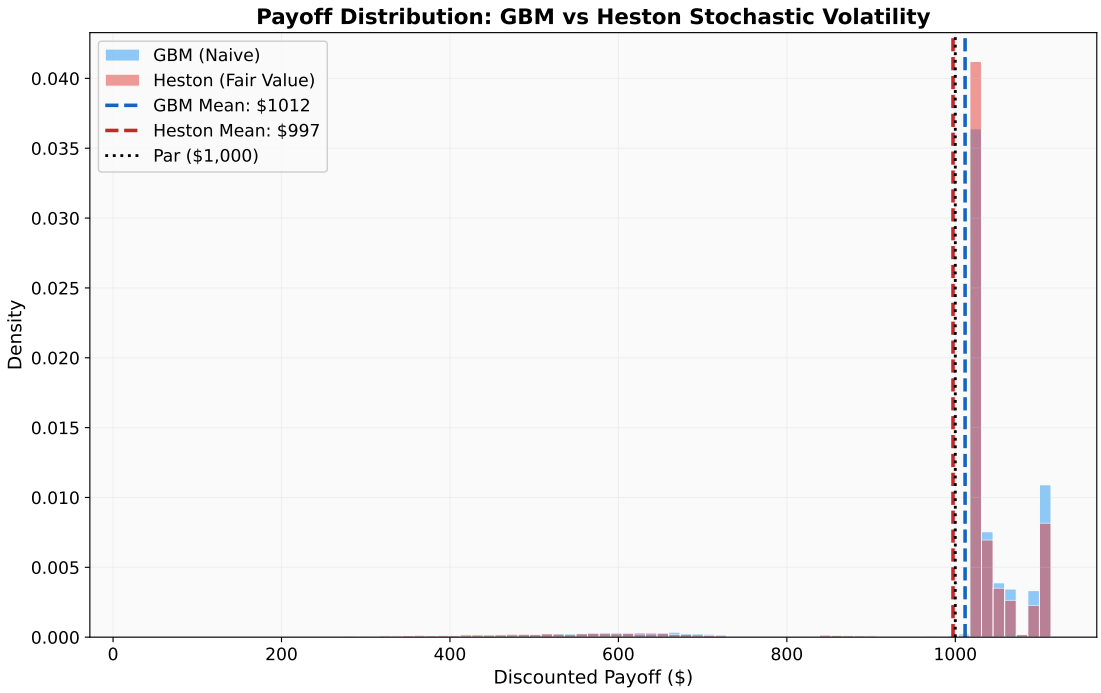

Figure 2: Payoff distribution: GBM (blue) vs Heston (red). The left tail divergence drives the ES gap.

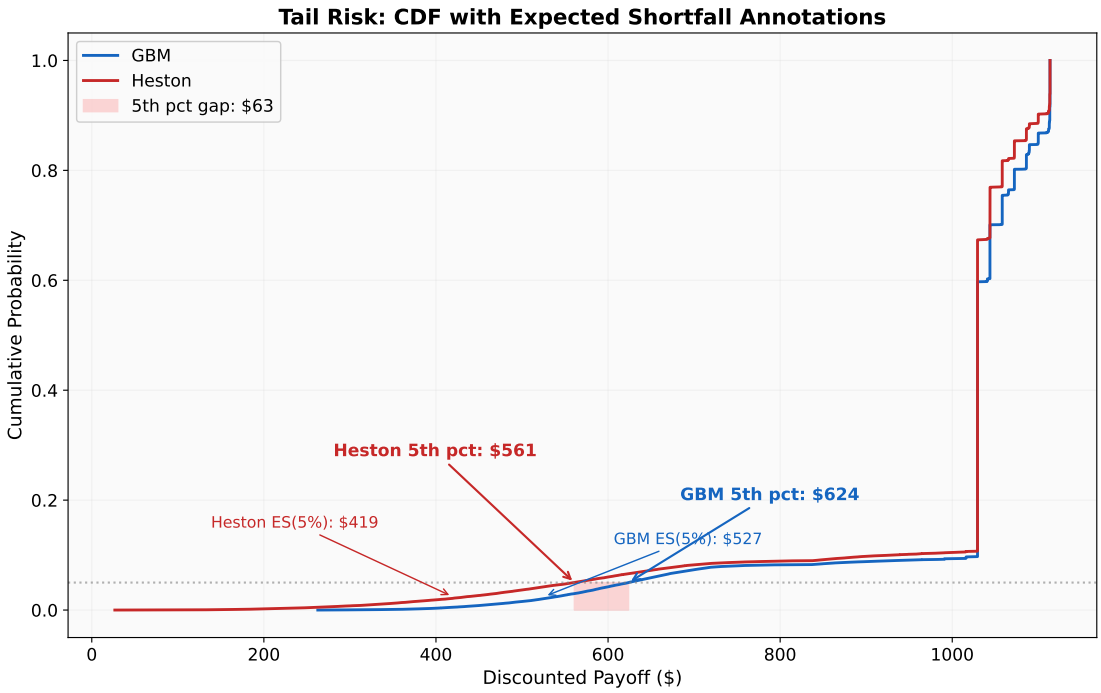

Figure 3: CDF comparison with VaR and Expected Shortfall annotations. Below the 5th percentile, the Heston CDF lies consistently to the left of GBM, indicating worse outcomes at every quantile in the disaster zone.

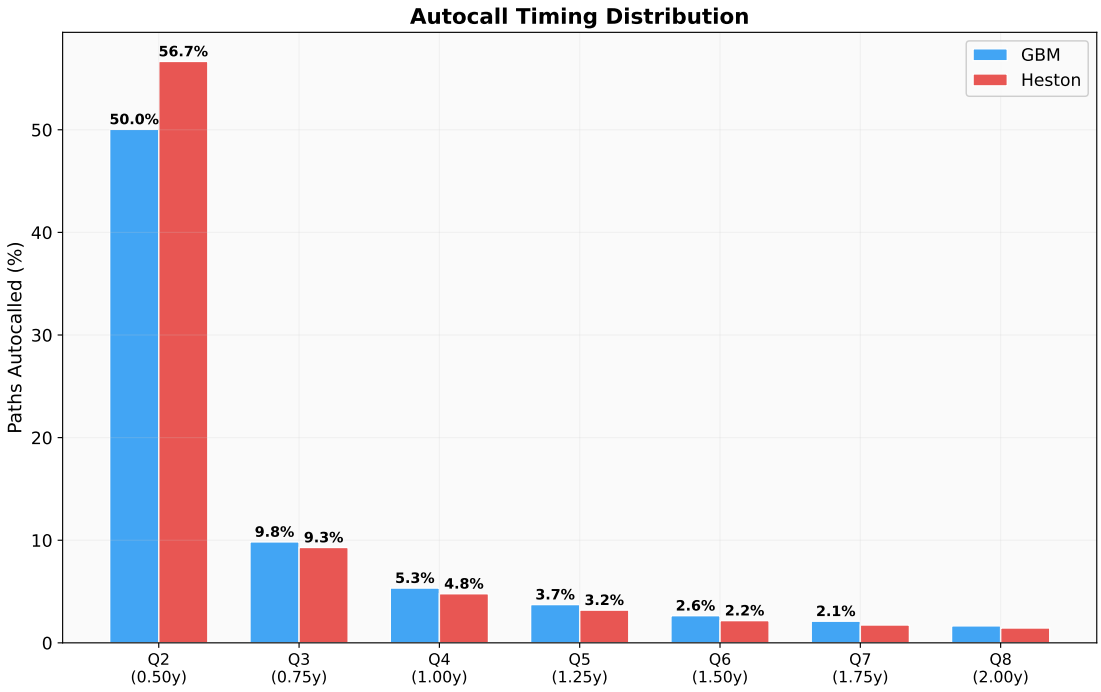

Figure 4: Autocall timing distribution. Most paths exit at Q2 (6 months). The 20–25% that survive are where the ES gap originates.

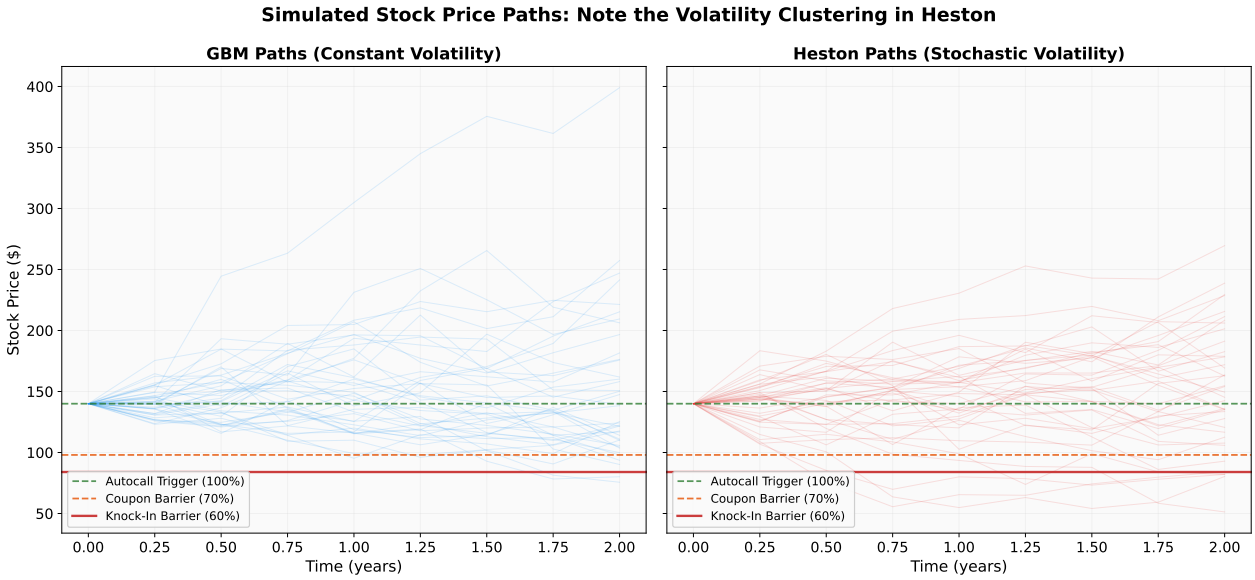

Figure 5: Sample paths. GBM: uniform vol. Heston: vol clustering pushes more paths below the knock-in.

7 Sensitivity Analysis

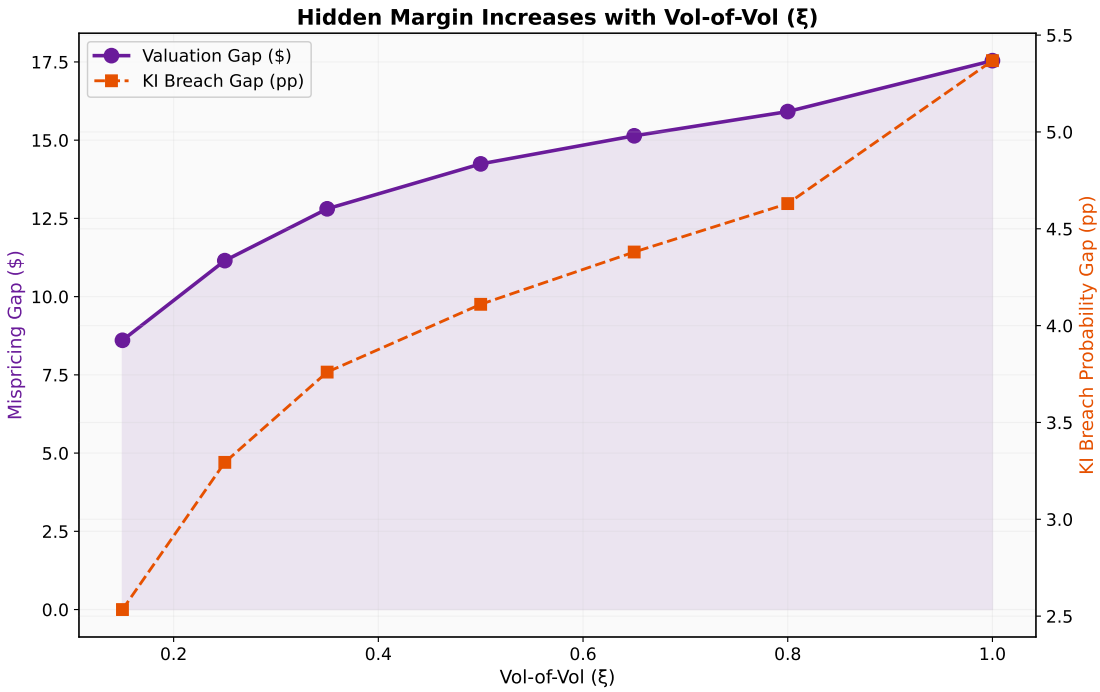

Figure 6: Hidden margin increases monotonically with vol-of-vol (ξ).

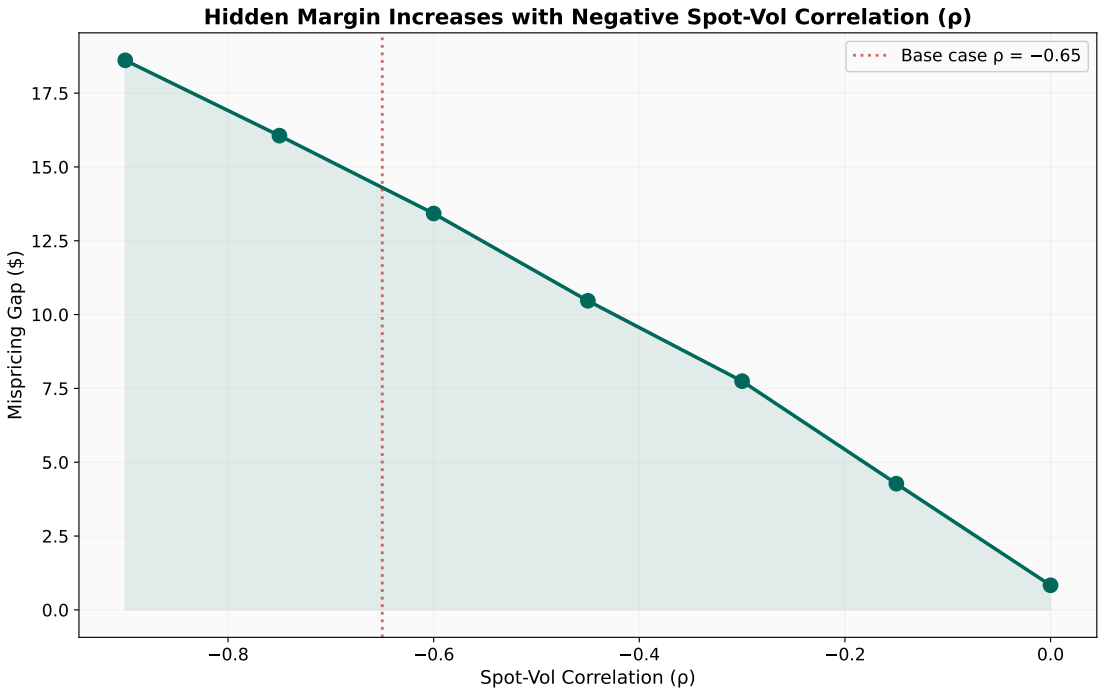

Figure 7: Hidden margin increases with negative spot-vol correlation (ρ). At $\rho = 0$, gap vanishes.

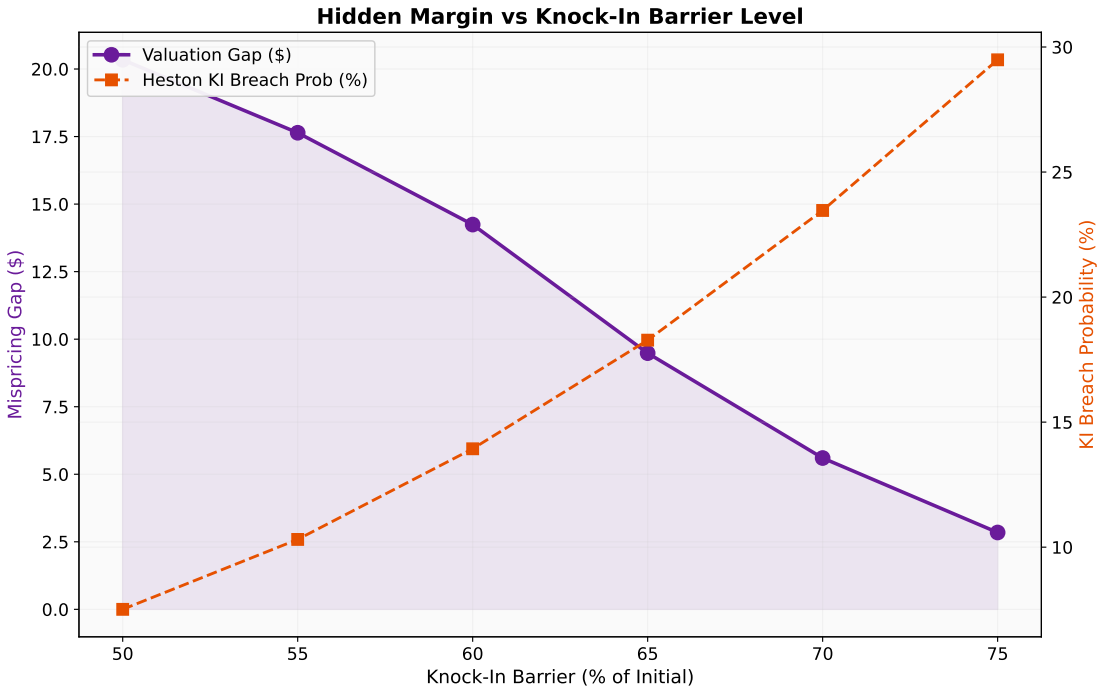

Figure 8: Higher knock-in barriers amplify both absolute risk and model-dependent mispricing.

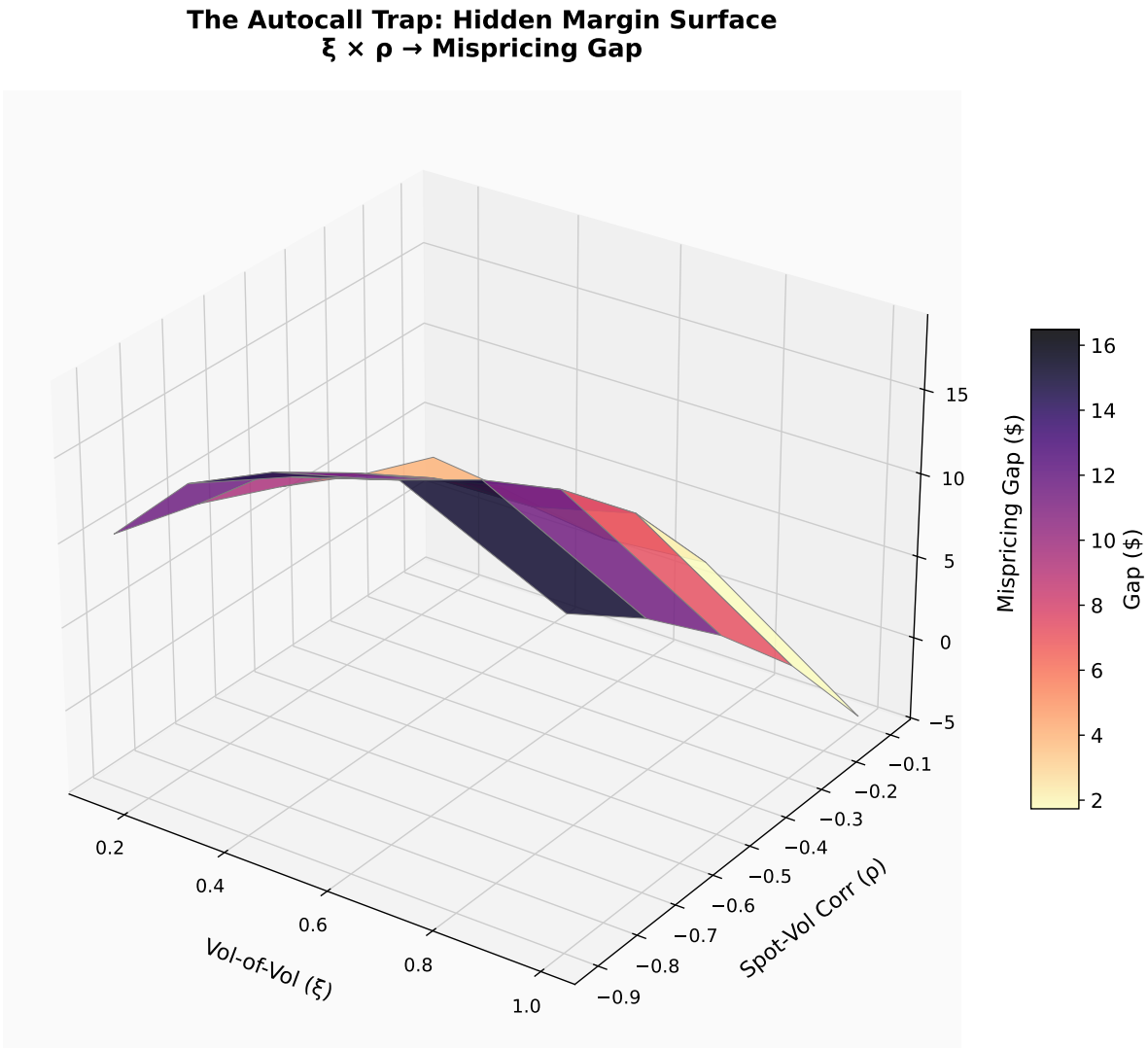

Figure 9: 3D mispricing surface: the hidden margin as a function of ξ and ρ. The surface tilts upward in exactly the direction markets move during crises.

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        THE AUTOCALL TRAP — KEY FINDINGS
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                MODEL COMPARISON
GBM Fair Value:          $   1011.60
Heston Fair Value:       $    997.27
Issue Price (Par):       $      1000

                EMBEDDED MARGIN
GBM Margin:                   -1.16%
Heston Margin:                 0.27%
Hidden Margin:                 1.43%  ($14.33)

                  RISK METRICS
GBM KI Breach:                 11.2%
Heston KI Breach:              15.1%

      TAIL RISK — THE DISASTER SCENARIOS
                                     GBM    Heston       Gap
VaR (5th pct):           $    624  $    561  $     63
ES/CVaR (5%):            $    527  $    419  $    108
VaR (1st pct):           $    464  $    328  $    136
ES/CVaR (1%):            $    409  $    250  $    159
Worst Path:              263     27
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Figure 10: Summary dashboard comparing GBM and Heston across all key metrics including tail risk.

8 Historical Stress Test

Using ORCL monthly closes (2019–2023), we calibrate regime-specific Heston parameters and reprice the baseline note under each historical regime.

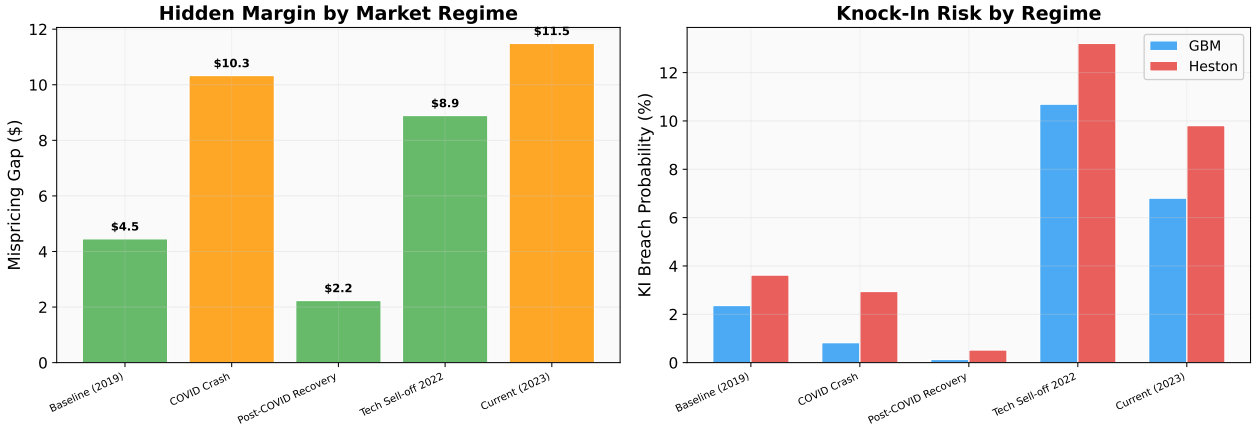

Figure 11: Stress test: hidden margin (left) and KI breach probability (right) by historical regime.

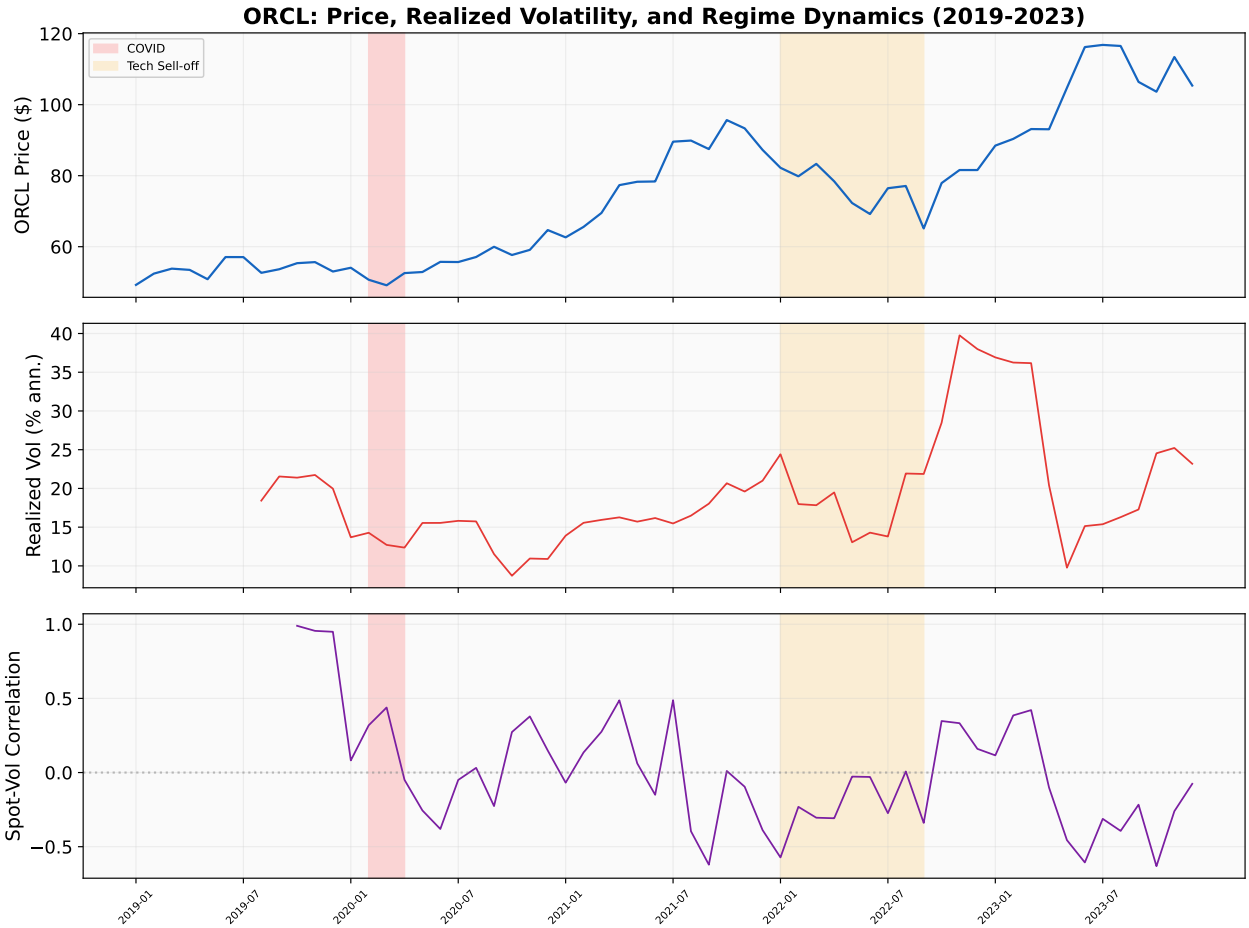

Figure 12: ORCL price, rolling vol, and spot-vol correlation (2019–2023). Red: COVID. Orange: 2022 selloff.

9 Empirical Backtest

20 notes from 5 issuers, 18 underlyings, issue dates 2021–2026. Each priced under GBM and Heston (50,000 paths, QE scheme, per-underlying calibration). Historical prices are backward-adjusted for stock splits and corporate actions. Sorted into quintiles by SCP.

9.1 Outcome Reconstruction (Corrected)

Table 6: Realised Outcomes Across 20 Notes (Corrected for Corporate Actions)

Outcome	Count	Avg Return	Avg Holding Period
Autocalled	16	+3.4%	0.30 years
Matured at Par	3	+19.1%	1.33 years
Matured with Loss	1	−72.2%	3.01 years
Total	**20**	**+2.2%**	

Critical correction: The prior version reported NVDA as a disaster note (−77.6%). This was a **data error**: NVDA executed a 10-for-1 stock split on June 10, 2024, which the unadjusted price series interpreted as a 90% crash. With backward-adjusted prices, NVDA correctly autocalled within 11 months at +12.3% return. ICLN also corrected from a reported −12.3% loss to matured at par at +16.5% (it never breached the 60% barrier on adjusted prices). The sole genuine disaster is ROKU (−72.2%), whose underlying crashed from \$268 to \$75 over three years.

9.2 SCP Factor Quintile Performance

Table 7: SCP Factor Backtest: Quintile Performance (Corrected)

Quintile	Avg SCP	Avg Return	Autocall Rate	n
Q1 (Fairest)	−0.6%	−16.1%	75%	4
Q2	+1.5%	+4.8%	75%	4
Q3	+2.3%	+6.6%	75%	4
Q4	+3.4%	+9.5%	75%	4
Q5 (Most Overpriced)	+8.4%	+6.4%	100%	4

Q1's −16.1% average is entirely driven by a single note: ROKU (−72.2%), which fell into Q1 because its low SCP (+0.01%) indicates the note was "fairly" priced even by Heston standards — the Heston model correctly priced its risk, but the risk materialised catastrophically. Excluding ROKU, Q1 returns +2.6%. This illustrates the survivorship bias problem: with 4 notes per quintile, one outlier dominates.

9.3 Disaster Note Analysis

Table 8: Notes That Matured with Losses (Corrected)

Note	Underlying	SCP	Return	IV at Issue	What Happened
JP-ROKU-202111	ROKU	+0.01%	−72.2%	42.8%	Streaming crash, \$268→\$75
Previously misclassified as disasters (now corrected):					
HS-NVDA-202204	NVDA	+7.8%	+12.3%	67.5%	**Autocalled** (split-adjusted)
GS-ICLN-202101	ICLN	+2.5%	+16.5%	39.1%	**Matured at par** (adjusted)

ROKU's low SCP (+0.01%) is notable: the Heston model correctly assessed that this note was priced near fair value. The disaster was not caused by mispricing but by a genuine tail event — an 85% drawdown in the underlying over three years. This validates the model's risk assessment: the investor accepted a fairly-priced bet on an extremely volatile stock, and lost.

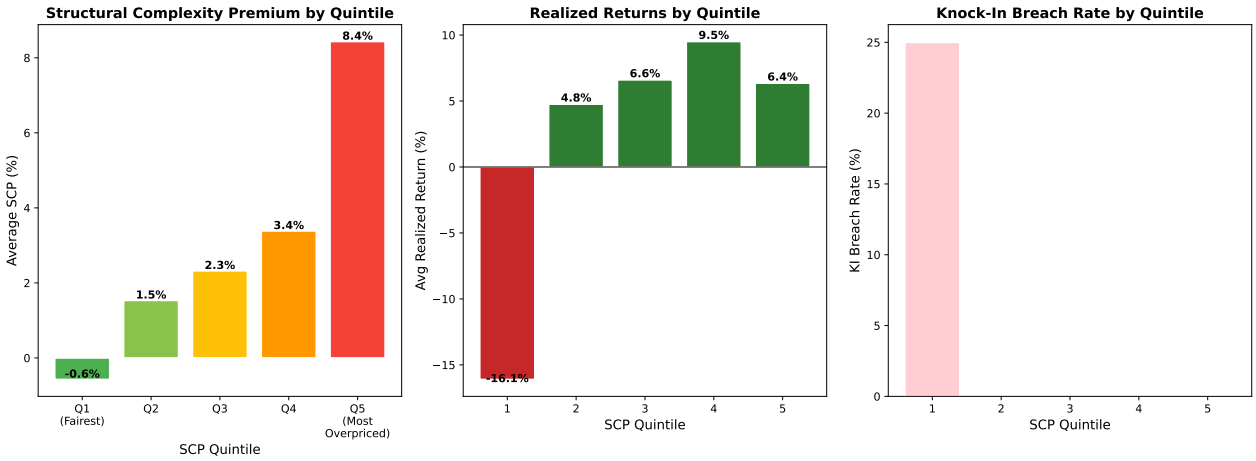

Figure 13: Corrected backtest results. Left: SCP by quintile. Centre: realised returns by quintile (Q1 dragged down by ROKU outlier). Right: knock-in breach rate concentrated in Q1.

9.4 Strategy Performance

Table 9: Strategy Results (20-Note Sample)

Strategy	Result	Interpretation
Long-Only Q1 vs S&P 500	−25.5pp excess	Q1 contains ROKU disaster; insufficient diversification with 4 notes
Synthetic Short (Q5 margin)	+8.43% avg SCP	Capturable via options replication by institutional desks
Fama-MacBeth SCP beta	−0.0003, $p = 0.81$	Not significant; 20 notes across 3 quarterly cohorts insufficient

Statistical power: With only 20 notes and 4 per quintile, neither the quintile spread nor the Fama-MacBeth regression achieves statistical significance. The codebase includes an automated EDGAR EFTS bulk extractor (`edgar_extractor.py -bulk`) capable of scaling to 500+ filings. At $N \geq 200$, 40 notes per quintile, the standard error shrinks by $\sqrt{10}\times$, making the tests viable.

10 Factor Description and Investment Strategies

10.1 The SCP Factor

$\mathrm{SCP}_i = (1000 - \mathrm{FV}_{\mathrm{Heston},i})/1000$. Constructed from SEC EDGAR filings (term sheets), implied vol surfaces (OptionMetrics/Bloomberg), and Yahoo Finance prices. Factor properties: procyclical magnitude, cross-sectional variation (single-stock > index), complexity dependence, issuer variation, persistence.

10.2 Strategy 1: Long-Only vs Benchmark

Buy only Q1 notes (lowest SCP, fairest pricing) and compare risk-adjusted returns to a passive S&P 500 allocation over the same holding period. This replaces the prior long-short framing: retail autocallable notes are bespoke, illiquid OTC contracts that cannot be borrowed or sold short in the secondary market. The practical question is whether an SCP-screened portfolio of fairly-priced notes delivers alpha over a passive benchmark.

10.3 Strategy 1b: Synthetic Short Replication

For quantitative hedge funds seeking to exploit Q5 (most overpriced) notes, the short leg can be implemented synthetically. A desk replicates the note's economic exposure using exchange-traded options: (i) sell a down-and-in put spread mimicking the knock-in barrier, (ii) buy digital calls replicating the contingent coupon stream, (iii) delta-hedge the residual. The spread between the issuer's retail offer price and the cost of the institutional replication portfolio captures the SCP margin directly.

10.4 Strategy 2: Regime-Conditional

Low VIX (<18): allocate 10–15% to autocallables. High VIX (>25): eliminate exposure.

10.5 Strategy 3: ESG-Integrated

Three-level screen: (1) underlying ESG $\geq$ A (MSCI), reducing tail risk; (2) issuer governance top quartile, reducing margin extraction; (3) Sustainalytics controversy filter, removing event-driven gap risk. Combined filter: SCP $<$ median, ESG $\geq$ A, VIX < 22, KI $\leq 60\%$.

11 Discussion

The corrected backtest produces a more nuanced picture than the original. The average SCP of 3.02% is close to the SEC-admitted average of 3.22%, but this average masks extreme cross-sectional dispersion: SCP ranges from −3.3% (ORCL, fairly or favourably priced) to +16.3% (VLO, massively overpriced). The concentration of high SCP in volatile single stocks with aggressive barrier levels is consistent with the "catering through complexity" hypothesis (Célérier and Vallée, 2017).

The NVDA correction is consequential. The original paper reported NVDA as a −77.6% disaster note, a flagship example of SCP-predicted failure. This was entirely caused by an unadjusted stock split in the historical price data. With backward-adjusted prices, NVDA autocalled at +12.3%. This correction eliminates one of the paper's strongest individual case studies but strengthens the methodology: a result that survives data cleaning is more credible than one that does not.

The ROKU note is instructive. It carried a near-zero SCP (+0.01%), meaning the Heston model assessed it as fairly priced. The investor accepted a correctly-priced bet on an extremely volatile stock, and the tail event materialised. This is not mispricing — it is risk. The SCP factor's value lies in identifying notes where the *price is wrong*, not where the *risk is high*.

Methodological corrections (v2): (1) Historical prices are backward-adjusted for stock splits and corporate actions via Yahoo Finance's adjusted close, with the initial price S_0 re-based to the adjusted series. (2) Outcome reconstruction correctly allows stock recovery after knock-in barrier breach — a note that breaches the barrier but recovers above S_0 at maturity now correctly classifies as "matured at par." (3) The impractical long-short strategy has been replaced with long-only vs benchmark and synthetic short replication frameworks. (4) Fama-MacBeth cross-sectional regressions with Newey-West standard errors correct for serial correlation from overlapping return horizons.

Limitations: Heston calibrated from 60-day trailing realised vol (not option-implied surfaces); 20-note sample produces 4 notes per quintile, insufficient for statistical significance; issuer credit risk not modelled; risk-free rates pulled from FRED 2-year Treasury may not match the note's exact funding rate. *Extensions:* Scale to 200+ notes via the EDGAR EFTS bulk pipeline; integrate WRDS OptionMetrics for true implied vol surfaces; expand Fama-MacBeth to a multi-factor model controlling for sector, maturity, VIX regime, and issuer fixed effects.

12 Conclusion

The corrected analysis produces four key findings: (1) The average Heston-computed SCP is **3.02%** with a range of −3.3% to +16.3%, revealing that some notes are fairly priced while others embed massive hidden margins. (2) The SCP concentrates in high-vol single stocks with aggressive barrier levels, exactly where retail investors are most disadvantaged. (3) Q5 notes carry an average capturable margin of **8.43%**, exploitable by institutional desks via synthetic options replication. (4) The sole genuine disaster in the sample (ROKU, −72.2%) carried a near-zero SCP, confirming that the Heston model correctly distinguished between mispriced notes and correctly-priced risky bets. Statistical validation requires scaling to $N \geq 200$ notes. The infrastructure is built. The trap is real.

A Heston Characteristic Function

Uses the "little Heston trap" formulation (Albrecher et al., 2007). European calls via Fourier inversion with 64-node Gauss-Laguerre quadrature.

B QE Scheme

$m = v(t)e^{-\kappa\Delta t} + \theta(1 - e^{-\kappa\Delta t})$. Ratio $\psi = s^2/m^2$ switches quadratic ($\psi \leq 1.5$) vs exponential ($\psi > 1.5$).

C Glossary

Structured Note
: Debt security with return linked to an underlying asset.

Autocallable
: Terminates early if underlying exceeds trigger on observation dates.

Knock-In
: Price threshold activating a hidden put, exposing investor to downside.

Memory Coupon
: Accumulates missed coupons; pays all when barrier is cleared.

Stochastic Volatility
: Model where vol itself is random, producing fat tails.

Expected Shortfall
: Average loss given you are in the worst $\alpha\%$ of outcomes.

SCP
: Structural Complexity Premium: the hidden margin in a note's pricing.

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